

Multi-channel neural vetting of exoplanet transit candidates: controlled ablations and distribution-free candidate selection

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Abstract

Transit surveys produce far more threshold-crossing events than planets, and deciding which is which is the bottleneck between photometry and a planet catalogue. We present PHANTOM, a vetting network that views each candidate through nine phase-folded representations — global and local flux, odd and even epochs, the secondary phase, two centroid channels, and folds at the competing period hypotheses $P/2$ and $2P$ — combined by a transformer over per-view tokens, together with a differentiable five-parameter transit model fitted by reconstruction alone. On the 15,683 labelled Kepler DR24 threshold-crossing events, PHANTOM reaches an average precision of 0.9732 ± 0.0097 against 0.9385 ± 0.0093 for a faithfully reimplemented ASTRONET trained on identical splits ($p = 4 \times 10^{-4}$, five seeds), and raises precision at 95 per cent recall from 0.805 to 0.927 — a 62 per cent reduction in the follow-up effort wasted on false positives.

We then subject the architecture to controlled ablation, and report two negative results that we believe are more useful than the headline. First, no individual component is responsible for the gain: removing the harmonic contrast head, the harmonic views, the differentiable decoder, the centroid channels or the catalogue diagnostics each costs at most 0.007 in average precision, and not one of the eight ablations is statistically significant. The advantage is collective and redundant, and single-component ablations therefore systematically understate it. Second, splitting the data by threshold-crossing event rather than by host star — a protocol we expected to leak —

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changes performance by 0.0002 ($p = 0.97$). We further show that gradient-boosted trees on summary statistics match PHANTOM on average precision (0.9754 ± 0.0056) while falling well short of it at high recall, that conformal p -values with Benjamini–Hochberg selection control the false discovery rate of the candidate list for target rates above about 5 per cent but not below, that the unsupervised decoder recovers transit durations and depths to 3.2 and 6.5 per cent of published values without ever seeing a parameter label, and that zero-shot transfer to 2,351 TESS objects of interest degrades to an average precision of 0.79 for both PHANTOM and ASTRONET, corroborating recent reports that cross-mission generalisation is not obtained for free.

Keywords: exoplanets, transit photometry, light curves, deep learning, conformal prediction, Kepler, TESS

1. Introduction

The transit method has produced the large majority of known exoplanets, but the detections it yields are not planets: they are *threshold-crossing events* (TCEs), periodic dimmings that survive a matched-filter search of a photometric time series. Most TCEs are not planetary. They arise from eclipsing binary stars, from the light of an eclipsing binary blended into the photometric aperture of an unrelated target, from stellar variability and starspots, and from instrumental systematics that happen to be periodic. The Kepler mission alone produced $\sim 3.4 \times 10^4$ TCEs in its final data release, of which only a minority correspond to planets (Thompson et al., 2018). Deciding which is which — *vetting* — is the bottleneck between a photometric survey and a planet catalogue.

Vetting has traditionally been performed by a cascade of physically motivated diagnostic tests. If odd-numbered and even-numbered transits have different depths, the signal is probably an eclipsing binary detected at half its true period. If a shallower dimming appears near the opposite orbital phase, the companion is self-luminous and therefore stellar. If the flux-weighted centroid of the target shifts during the event, the light is being removed from a neighbouring source rather than from the target. If the profile is V-shaped rather than flat-bottomed, the eclipse is probably grazing. These tests are fast, interpretable, and were the basis of the Kepler Robovetter (Thompson et al., 2018). They are also brittle: each is a hand-tuned threshold on a statistic that degrades rapidly as the signal-to-noise ratio falls, and each

fails in a characteristic way. The odd-even test, in particular, is defeated by exactly the situation it is meant to catch, because a binary detected at its half-period alias presents both eclipses as “primary” events.

Deep learning changed the picture. Shallue and Vanderburg (2018) showed that a one-dimensional convolutional network applied to two fixed-length representations of the phase-folded light curve — a whole-orbit “global” view and a zoomed “local” view — outperformed the hand-built cascade, and used it to recover two previously missed planets. Subsequent work added input channels rather than changing the reasoning: Ansdell et al. (2018) supplied centroid time series and stellar parameters, and EXOMINER (Valizadegan et al., 2022) added a branch per diagnostic test and validated 301 new planets. The same recipe was carried to TESS (Yu et al., 2019; Osborn et al., 2020), and more recently attention has been introduced over the same two views, together with post-hoc temperature scaling, by Islam (2026).

Two concurrent studies bear directly on the present work. Priyanshu (2026) give the first application of conformal prediction to transit *detection*, using a self-supervised stellar world model to find single transits in raw flux and reporting empirical coverage of conformal prediction sets. Kopparapu et al. (2026) examine cross-instrument prioritisation over all train/test combinations of Kepler and TESS, and find that a model trained on one mission transfers poorly to the other, recommending joint training instead. We return to both in Section 7: our conformal contribution targets a different guarantee (false discovery rate over a selected list, rather than coverage of a prediction set) on a different task (vetting events of known period, rather than detecting single transits), and our cross-mission experiment is deliberately zero-shot, which is the setting Kopparapu et al. (2026) identify as difficult.

Across this line of work the network is given the folded light curve *at the period reported by the search* and asked to classify what it sees. That framing embeds an assumption which the classical tests explicitly do not make: that the reported period is the true period. It is precisely the failure of that assumption — the half-period alias — which produces the dominant astrophysical false positive. A network shown only the fold at P can learn proxies for the alias, but it cannot examine the alternative hypothesis directly, because the evidence that distinguishes P from $2P$ lives in a fold the network is never shown.

A second gap concerns what the network outputs. Vetting produces a ranked list that consumes finite follow-up resources, so the operationally meaningful question is not “what is the score of this candidate” but “if I

observe everything on this list, what fraction of my time is wasted”. A softmax output does not answer that question: neural network probabilities are systematically miscalibrated (Guo et al., 2017), and calibration on one data set carries no guarantee on another. Nor do these classifiers estimate the physical parameters — period, depth, duration — that the survey exists to measure; those are taken from the upstream search and passed through unchanged.

This paper presents PHANTOM, a vetting network built around those two gaps, and — more importantly — a controlled study of which of its parts actually earn their place. Our contributions are as follows.

1. **A multi-channel vetting architecture.** Nine phase-folded views, including folds at the competing period hypotheses $P/2$ and $2P$, are encoded independently and combined by a transformer over per-view tokens carrying view-type and period-hypothesis embeddings (Section 4.2). On the standard Kepler DR24 benchmark this improves average precision over a faithfully reimplemented ASTRONET by 0.035 at matched training budget and slightly *lower* parameter count, and cuts the false-positive content of a 95 per cent recall candidate list by 62 per cent.
2. **A differentiable transit decoder.** A five-parameter analytic transit model is rendered onto the local view and fitted by reconstruction with no parameter supervision (Section 4.3). Its bottleneck is directly interpretable: the ingress-softness term separates flat-bottomed planetary profiles from V-shaped stellar eclipses, and the fitted depths and durations agree with published values to 6.5 and 3.2 per cent respectively without ever seeing a parameter label.
3. **Distribution-free candidate lists.** Conformal p -values computed against held-out known non-planets, combined with the Benjamini–Hochberg procedure, target a candidate list whose false discovery rate is controlled at a chosen level under an exchangeability assumption alone (Section 5.3). We verify empirically where this works and where it fails.
4. **Two negative results.** We ablate every component over five seeds and find that *none* of them individually accounts for the gain: the largest single effect is 0.007 in average precision and no ablation reaches significance. We also find that the star-level split we adopted to prevent leakage makes no measurable difference relative to the event-level split used by earlier work. We report both in full, because a field that

publishes only the components that worked will keep re-discovering the ones that did not.

We regard the second pair as the more useful contribution. Individually plausible physical priors — harmonic folds, an explicit alias contrast, a differentiable transit model, centroid channels — are each individually dispensable here, because the evidence they carry is largely redundant with what the remaining channels already supply. That is a property of the problem rather than of any one implementation, and it has a direct methodological consequence: one-component-at-a-time ablation, the standard instrument for attributing credit in this literature, systematically understates the value of a redundant ensemble of evidence.

2. Data

2.1. *The Kepler DR24 threshold-crossing event set*

We use the Kepler Q1–Q17 DR24 TCE catalogue hosted at the NASA Exoplanet Archive, which is the benchmark introduced by Shallue and Vanderburg (2018) and reused by Ansdell et al. (2018). Of its 20,367 TCEs, 15,737 carry a disposition from the Autovetter training set (McCauliff et al., 2015):

- **PC** (planet candidate, 3,600 events) — the positive class;
- **AFP** (astrophysical false positive, 9,596 events) — eclipsing binaries, blends and other genuine astrophysical signals that are not planetary transits of the target;
- **NTP** (non-transiting phenomenon, 2,541 events) — stellar variability and instrumental artefacts.

These 15,737 events lie on 9,865 distinct stars; 15,683 of them survive preprocessing (Section 3) and constitute the working set. Figure 1 shows the nine input channels for one event of each class. We treat PC as the positive class and {AFP, NTP} as the negative class, following the convention of the prior work we compare against.

Light curves are the long-cadence (29.4 min) PDCSAP photometry from the Kepler pipeline, retrieved for every quarter in which the target was observed. We retain cadences whose quality flag contains none of the bits indicating attitude tweaks, safe modes, coarse or Earth point, desaturation events, manual exclusions or detector anomalies.

2.2. The TESS cross-mission set

To test whether the learned representation is specific to Kepler we assemble a zero-shot evaluation set from the TESS Objects of Interest catalogue (Guerrero et al., 2021). We keep TOIs whose TFOPWG disposition is unambiguous — confirmed or known planets (CP, KP) as positives, false positives and false alarms (FP, FA) as negatives — and which have a published epoch, period and duration. Photometry is the two-minute-cadence SPOC light curve, limited to the first four sectors available for each target; only cadences with a zero quality flag are kept, since the TESS quality bits differ from Kepler’s.

3. From photometry to fixed-length views

3.1. Detrending

Stellar variability and residual instrumental trends exceed a planetary transit in amplitude by orders of magnitude, so each light curve is flattened before folding. Following Vanderburg and Johnson (2014) and Shallue and Vanderburg (2018) we divide each contiguous segment — data separated by gaps longer than 0.75 d are treated as separate segments — by a cubic basis spline fitted with iterative 3σ outlier rejection.

Two details matter. First, the knot spacing is not fixed: it is selected per target from a logarithmic grid spanning 0.5 to 20 d by minimising the Bayesian Information Criterion summed over all segments, so that a rapidly rotating star and a photometrically quiet one are treated differently. Second, the cadences within 1.5 transit durations of *any* TCE on the target are excluded from the spline fit, though the fitted spline is still evaluated at those cadences. Without this mask the spline partially absorbs the very signal being measured, biasing depths low.

The centroid time series are detrended in the same way and with the same knot spacing, so that only excursions coincident with an event survive.

3.2. Views

For each TCE, cadences falling within 1.5 durations of a *different* TCE on the same star are removed, and the remainder is folded on the reported ephemeris. Nine fixed-length views are then rendered by median-binning (Table 1). Each is centred on its median and divided by the absolute value of its minimum, so that the deepest bin sits at -1 ; the divisor is retained

Table 1: The nine input views. The period hypothesis column indicates which trial period the view is folded on; these labels are supplied to the network as embeddings.

View	Bins	Hypothesis	Purpose
Global	2001	P	whole-orbit morphology
Local	201	P	transit shape
Odd	201	P	depth on odd epochs
Even	201	P	depth on even epochs
Secondary	201	P	occultation by a luminous companion
Harmonic	201	$P/2$	is the true period half of the reported one?
Harmonic	2001	$2P$	does the fold separate into unequal eclipses?
Centroid (global)	2001	P	photocentre motion, whole orbit
Centroid (local)	201	P	photocentre motion during the event

as a scalar, which is what allows an absolute depth to be recovered from the normalised view in Section 4.3.

The two harmonic views are the novel channels. Folding at $2P$ maps alternating eclipses of a binary onto *distinct* phases, so a depth difference that the odd-even test must infer from two interleaved subsets becomes directly visible as two separated dimmings of unequal depth. Folding at $P/2$ tests the opposite hypothesis: if the true period is half the reported one, that fold sharpens the event instead of smearing it.

The local, odd, even and secondary views span ± 4 transit durations. The secondary view is centred by re-folding about the epoch of the deepest out-of-primary bin, rather than by windowing inside the primary fold, so that the window remains fully populated when the secondary lies near phase ± 0.5 .

3.3. Scalar features

Alongside the views the network receives scalar features in four groups: *transit* (period, duration, depth, model signal-to-noise, multiple event statistic, number of transits, impact parameter, radius and semi-major axis ratios), *stellar* (effective temperature, surface gravity, metallicity, inferred planet radius), *diagnostic* (the pipeline’s odd-even and centroid offset statistics and robust fit statistic), and *derived* — eleven quantities measured here from the detrended light curve, including the odd-even depth asymmetry, the secondary depth ratio, the selected knot spacing, the out-of-transit scatter, the normalisation scale, and a red-noise proxy formed from the ratio of the scat-

Input channels: a planet candidate (PC), an astrophysical false positive (AFP) and a non-transiting phenomenon (NTP)

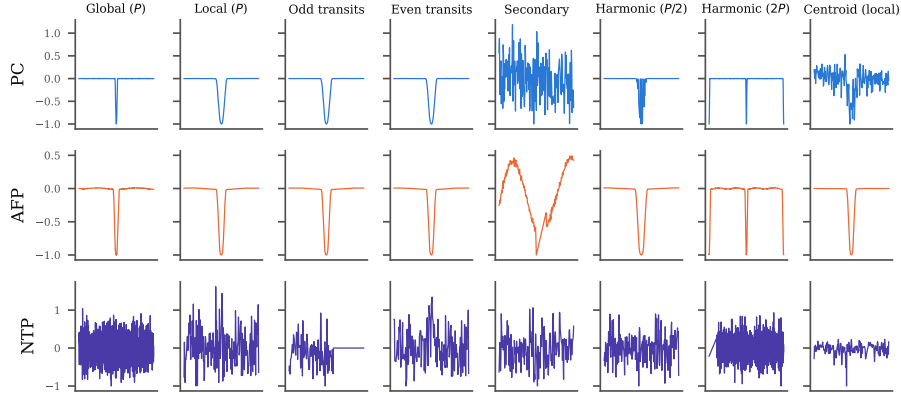


Figure 1: The nine input channels for a planet candidate (PC), an astrophysical false positive (AFP) and a non-transiting phenomenon (NTP). The planet shows a flat-bottomed transit with matched odd and even depths and a featureless secondary phase; folding it at $2P$ correctly separates alternate transits.

ter of six-hour bins to the white-noise expectation. The derived group is the only one computable identically for both missions and is therefore the configuration used for the cross-mission experiment. Features spanning orders of magnitude are log-compressed, missing values are imputed to the training-split median, and all are standardised with training-split statistics.

4. The PHANTOM architecture

PHANTOM has three parts: a per-view convolutional encoder, a harmonic attention block over the resulting tokens, and a differentiable transit decoder.

4.1. View encoders

Each view is encoded by a stack of pre-activation residual blocks with GroupNorm, interleaved with strided max-pooling — five stages for the 2001-bin views and three for the 201-bin views. The final feature map is reduced by concatenated mean and max pooling, capturing respectively the average level and the depth of the deepest excursion, and projected to a token of dimension $d = 256$. Views of equal length share an architecture but not weights, since a centroid view and a flux view carry different meaning.

4.2. Harmonic-hypothesis attention

Each token receives two additive embeddings: a view-type embedding and a *period-hypothesis* embedding drawn from $\{P/2, P, 2P\}$. A learnable classification token is prepended and the sequence is processed by a pre-norm transformer encoder (Vaswani et al., 2017) of three layers and eight heads.

Attention alone leaves the harmonic comparison implicit, so we add an explicit contrast. Writing $h_{P/2}$, h_P and h_{2P} for the mean of the encoded tokens under each hypothesis, the contrast vector

$$c = [h_P \parallel h_P - h_{P/2} \parallel h_P - h_{2P} \parallel |h_P - h_{2P}|] \quad (1)$$

is passed through a single hidden layer and concatenated into the classifier input. The difference terms state directly whether the evidence prefers the reported period over its harmonics; the absolute-value term makes the comparison insensitive to the sign of the discrepancy, which is what the eclipsing-binary hypothesis requires.

4.3. Differentiable transit decoder

Let x index the local-view grid in units of transit durations from mid-transit. From the encoded local-view token an affine head predicts five bounded parameters: depth $\delta > 0$, half-duration $T \in (0.05, 2)$, phase offset $x_0 \in (-0.5, 0.5)$, limb-darkening strength $u \in (0, 1)$ and ingress softness $s \in (0.02, 1)$. With $z = |x - x_0|/T$ the model light curve is

$$\hat{f}(x) = -\delta \underbrace{\sigma\left(\frac{1-z}{s}\right)}_{\text{in-transit indicator}} \underbrace{\left[1 - u(1 - \sqrt{\max(1 - z^2, 0)})\right]}_{\text{limb darkening}}, \quad (2)$$

where σ is the logistic function. Equation 2 is analytic and differentiable in all five parameters, so it can be trained by backpropagation through a reconstruction loss against the observed local view.

The softness s is the physically interesting parameter. As $s \rightarrow 0$ the indicator approaches a step and the profile becomes flat-bottomed, the signature of a small opaque body crossing the stellar disc. As s grows the ingress and egress ramps broaden until the profile is V-shaped, the signature of a grazing or blended stellar eclipse. A single scalar therefore expresses the U-versus-V distinction that the classical shape test tries to capture with a ratio of widths, but it is fitted jointly with depth, duration and limb darkening rather than measured on a noisy curve in isolation.

Two quantities derived from the fit enter the classifier: the root-mean-square reconstruction residual and its maximum absolute value. A genuine transit is well described by Equation 2; a stellar eclipse or an instrumental artefact is not, and the residual records that mismatch.

Critically, the five parameters are *not* regressed against catalogue values. Supervising them would teach the network to reproduce the pipeline fit against which it is later compared, making any parameter-recovery result circular. They are identified by reconstruction alone.

4.4. Objective

The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(\hat{y}, y) + \lambda \|\hat{f} - f_{\text{local}}\|_2^2, \quad (3)$$

with $\lambda = 5$. Networks are optimised with AdamW (Loshchilov and Hutter, 2019) under a one-cycle schedule, batch size 128, gradient clipping at norm 5, and early stopping on validation average precision. Views are randomly mirrored about mid-transit during training, which is a valid symmetry of a transit profile.

5. Evaluation protocol

5.1. Splits

Our primary protocol splits 80/10/10 into training, validation and test sets *by star*: every TCE on a given target falls in exactly one fold. Multiple TCEs on one star share a light curve, the same detrending solution, the same stellar parameters and the same instrumental systematics, so a TCE-level split places near-duplicates on both sides of the boundary and inflates test scores. Because previous work reports the TCE-level split, we also report it, so that the comparison to published numbers is like-for-like and the size of the leakage is itself measurable.

5.2. Metrics

We report the area under the ROC curve, average precision (the area under the precision-recall curve), and precision at fixed recall. The last is the operational quantity: a vetting pipeline is run at high recall so as not to discard planets, and is then judged by how much follow-up effort the surviving false positives consume. All figures are averaged over five random seeds and reported with their standard deviation; ensembles average the per-seed probabilities.

5.3. Distribution-free candidate selection

Let s denote the model score. Treating “this event is not a planet” as the null hypothesis, and given n scores $s_1, \dots, s_n$ of *known non-planets* from the held-out validation split, the conformal p -value of a test event with score s is

$$p = \frac{1 + \#\{i : s_i \geq s\}}{n + 1}. \quad (4)$$

Applying the Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995) at level q to these p -values yields a candidate list whose false discovery rate is controlled at q (Bates et al., 2023). The guarantee is finite-sample and distribution-free: it requires only that calibration and test events be exchangeable, and holds regardless of whether the network is well specified or well calibrated. For comparison we also report the expected calibration error before and after temperature scaling (Guo et al., 2017).

5.4. Transit parameter recovery

The decoder bottleneck is converted to physical units by $\delta_{\text{ppm}} = \delta \kappa \times 10^6$, where κ is the normalisation scale divided out of the local view, and $T_{\text{dur}} = 2T T_{\text{cat}}$, where T_{cat} is the catalogue duration defining the grid. Uncertainties are the standard deviation across ensemble members (Lakshminarayanan et al., 2017).

These estimates are compared against transit parameters from the NASA Exoplanet Archive’s composite planetary parameters table — that is, against published follow-up analyses rather than against the Kepler pipeline whose fit is being assessed. The comparison is therefore between two independent estimates of the same physical quantity: the pipeline value that seeded the search, and the value recovered here.

6. Results

All numbers are test-set values on the star-level split, averaged over five random seeds and quoted with their standard deviation; significance is assessed with Welch’s t -test on the per-seed values.

6.1. Comparison with baselines

Table 2 gives the headline comparison. PHANTOM improves average precision over ASTRONET by 0.0347 ($t = 5.75$, $p = 4.3 \times 10^{-4}$) while using

Table 2: Test-set performance on the 15,683 labelled Kepler DR24 threshold-crossing events, star-level split, five seeds. P@R denotes precision at the stated recall.

Model	Params	AUC	AP	P@R= 0.90	P@R= 0.95
Classical cascade	–	0.7668 ± 0.0162	0.4017 ± 0.0193	0.271	0.271
ASTRONET	10.55 M	0.9825 ± 0.0028	0.9385 ± 0.0093	0.868	0.805
GBDT (scalars)	–	0.9923 ± 0.0025	0.9754 ± 0.0056	0.939	0.902
PHANTOM	9.16 M	0.9929 ± 0.0027	0.9732 ± 0.0097	0.962	0.927

slightly fewer parameters (9.16M against 10.55M), so the difference is attributable to the architecture and the input channels rather than to capacity. The reimplemented ASTRONET reaches AUC 0.9825 here; the originally published value is 0.988, and the small shortfall is expected given that our preprocessing chain, quality masking and split protocol differ in detail from the original.

The operationally meaningful comparison is the last column. A vetting pipeline is run at high recall so as not to discard planets, and is then judged by the follow-up effort its false positives consume. At 95 per cent recall the ASTRONET candidate list is 19.5 per cent false positives and the PHANTOM list is 7.3 per cent, a reduction of 62 per cent in wasted effort. Figure 2 shows the full precision-recall and ROC curves.

Two baseline results deserve emphasis. The classical cascade — the odd-even, secondary-eclipse and V-shape tests, implemented carefully and with the look-elsewhere correction that the secondary test requires — reaches an average precision of only 0.40. Its individual tests are nonetheless sharply diagnostic: the corrected secondary test fires on 34 per cent of astrophysical false positives and 1 per cent of planet candidates. The gap between sharp individual tests and poor aggregate performance is precisely the gap that learned combination closes.

More surprisingly, gradient-boosted trees on the scalar feature vector match PHANTOM on average precision (0.9754 versus 0.9732, $p = 0.67$) and beat ASTRONET comfortably, despite never seeing a light curve. The scalar vector includes the pipeline’s own data-validation diagnostics, which encode a great deal of the vetting problem already. The tree nevertheless falls behind at high recall (0.902 against 0.927 at 95 per cent recall), which is where a survey operates. Any claim that a deep model is necessary for this task must clear this baseline, and we would encourage its inclusion in future work.

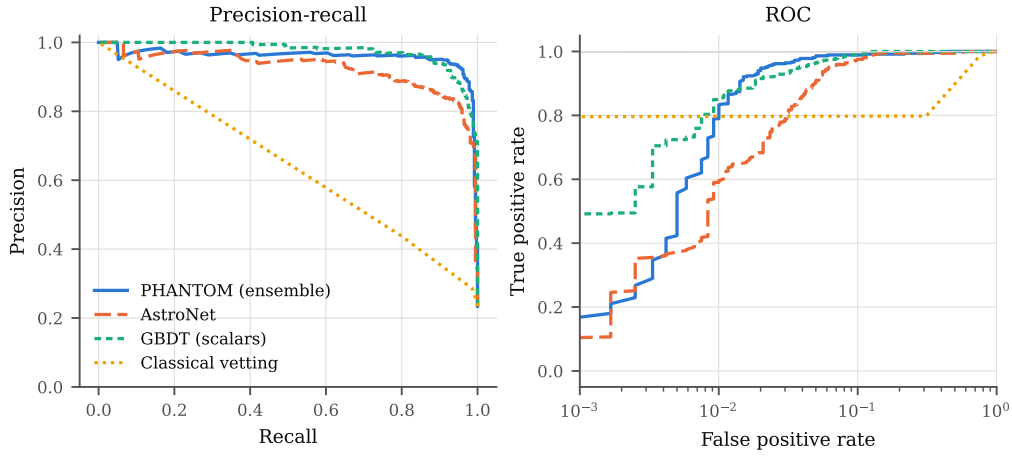


Figure 2: Precision-recall and ROC curves on the held-out test set. The ROC abscissa is logarithmic to resolve the low-false-positive regime.

6.2. The split protocol makes no difference

We adopted a star-level split on the expectation that multiple TCEs on one star — sharing a light curve, a detrending solution, stellar parameters and systematics — would leak between folds under the event-level split used by earlier work. They do not. PHANTOM scores 0.9732 ± 0.0097 under the star-level split and 0.9734 ± 0.0059 under the event-level split ($p = 0.97$); ASTRONET scores 0.9385 and 0.9315 ($p = 0.19$). The effect is consistent with zero for both models.

This is not because sharing is rare. 65.1 per cent of the labelled events sit on a star carrying at least one other labelled event, and 44.3 per cent of the stars host more than one. The explanation must therefore be that sibling events are not near-duplicates of each other. They are typically dissimilar signals — a genuine transit alongside a systematic at an unrelated period, with different depths, durations and shapes — so knowing the label of one conveys little about the other beyond the shared stellar and instrumental context, which is apparently not what the classifier keys on.

We continue to recommend the star-level split on principle, since its cost is nil and the argument for it does not depend on the size of the effect. But we can report that published results using the event-level split are not measurably inflated by it, and that the leakage concern, at least on this benchmark, is unfounded.

Table 3: Component ablations, five seeds each, star-level split. Δ AP is the full model minus the ablated model, so positive values indicate a component that helps. No row is significant at the 5 per cent level.

Configuration	AP	Δ AP	p
PHANTOM (full)	0.9732 ± 0.0097	—	—
– harmonic contrast head	0.9738 ± 0.0065	-0.0007	0.90
– harmonic views ($P/2$, $2P$)	0.9702 ± 0.0119	$+0.0030$	0.68
– differentiable decoder	0.9690 ± 0.0061	$+0.0042$	0.44
– catalogue diagnostics	0.9670 ± 0.0095	$+0.0062$	0.34
– centroid views	0.9660 ± 0.0109	$+0.0072$	0.31
global and local views only	0.9627 ± 0.0117	$+0.0104$	0.16
light-curve-derived scalars only	0.9621 ± 0.0113	$+0.0110$	0.14
no scalar inputs	0.9610 ± 0.0168	$+0.0121$	0.21
ASTRONET (different architecture)	0.9385 ± 0.0093	$+0.0347$	4×10^{-4}

6.3. Ablations: no component is individually responsible

Table 3 removes each component in turn. The result is unambiguous and, to us, unexpected: *not one of the eight ablations is statistically significant*, and the largest single effect is 0.0072 in average precision.

The pattern is nonetheless orderly. Effects grow monotonically as more evidence is withdrawn: removing one channel costs little, removing all the extra channels costs 0.0104, removing all scalars costs 0.0121, and only when the architecture is changed as well does the full 0.0347 gap to ASTRONET open up. The components are individually dispensable and collectively necessary, which is the signature of redundant evidence. A grazing eclipsing binary, for instance, betrays itself in the V-shaped local profile, in the decoder’s ingress-softness parameter, in the odd-even depth ratio and in the $2P$ fold; deleting any one leaves the other three.

This has a methodological consequence worth stating plainly. Leave-one-out ablation is the standard instrument for attributing credit in this literature, and under redundancy it is close to uninformative — it will report that nothing matters even when the ensemble of components clearly does. The comparison that carries information is the nested sequence, not the individual deletion.

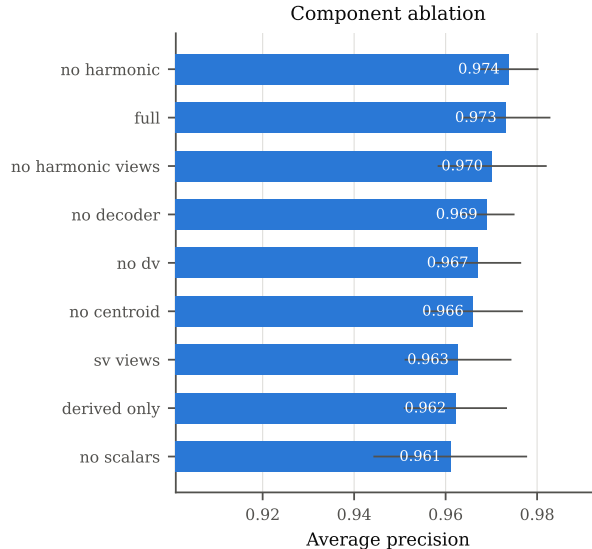


Figure 3: Component ablations with five-seed error bars. The overlap of every interval with the full model is the central negative result of this study.

6.4. Calibration and distribution-free candidate selection

A five-member ensemble on a fixed split improves average precision by 0.0028 and precision at 95 per cent recall by 0.0113 over its individual members, and reduces the expected calibration error from 0.0142 to 0.0120 under temperature scaling ($T = 1.12$). The network is already close to calibrated before scaling, which is not typical of deep classifiers (Guo et al., 2017) and which we attribute to the modest capacity and strong augmentation.

Calibration, however, is a statement about average agreement, not a guarantee. Table 4 reports the realised false discovery rate of the conformal selection of Section 5.3 against its nominal target. For targets of 5 per cent and above the procedure behaves as advertised: at $q = 0.10$ it selects 389 candidates with a realised FDR of 0.090 and 96.7 per cent recall, and at $q = 0.20$ it realises 0.194. Below 5 per cent it does not: at $q = 0.01$ the realised rate is 0.033, more than three times nominal.

The failure is expected rather than anomalous, and has two causes. The calibration set contains roughly 1,200 negatives, so the smallest attainable conformal p -value is $\sim 8 \times 10^{-4}$ and the discrete p -value grid is coarse exactly where small q needs resolution. More fundamentally, conformal p -values computed against a shared calibration set are mutually dependent,

Table 4: Conformal candidate selection with Benjamini–Hochberg at level q , calibrated on the validation-split negatives. Control holds for $q \gtrsim 0.05$ and fails below it.

Target q	Selected	Realised FDR	Recall
0.01	181	0.033	0.478
0.02	290	0.034	0.765
0.05	353	0.057	0.910
0.10	389	0.090	0.967
0.20	449	0.194	0.989

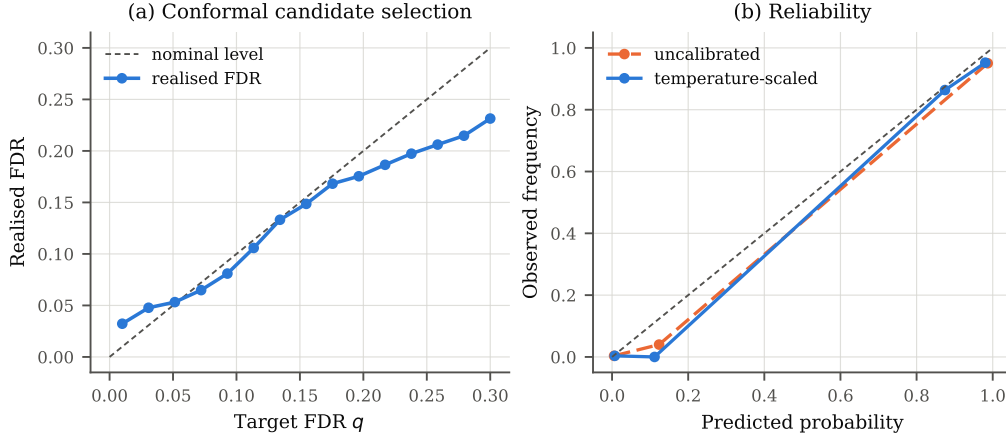


Figure 4: (a) Realised against nominal false discovery rate for the conformal selection; control holds above $q \simeq 0.05$ and fails below it. (b) Reliability diagram before and after temperature scaling.

and Benjamini–Hochberg controls the false discovery rate under such dependence only asymptotically unless the conditional-calibration variant of Bates et al. (2023) is used. Practitioners wanting guarantees at the one-per-cent level should enlarge the calibration set or adopt that variant; at the five-to-twenty per cent level that a follow-up programme actually operates in, the simple procedure is adequate.

6.5. Transit parameters from an unsupervised decoder

The decoder’s five parameters are never supervised. Converted to physical units and compared against published values for the 264 confirmed planets in the test split, the recovered transit durations have a median absolute

Table 5: Transit parameter recovery against published literature values for the 264 confirmed planets in the test split. Median absolute relative error; the fraction of systems recovered to within 10 per cent is given in parentheses.

Estimator	Duration	Depth
Kepler DV catalogue	2.0% (90%)	3.3% (86%)
PHANTOM decoder	3.2% (89%)	6.5% (66%)

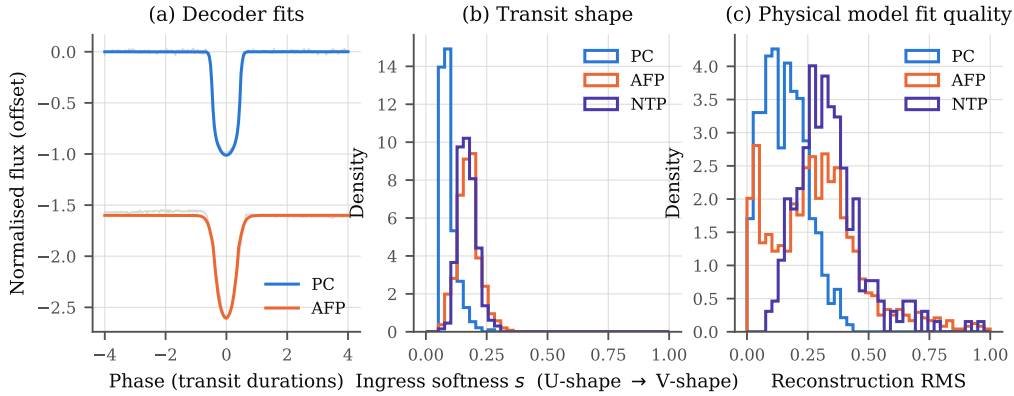


Figure 5: The differentiable decoder. (a) Fitted profiles against observed local views. (b) The ingress-softness parameter s separates flat-bottomed planetary transits from V-shaped false positives. (c) Reconstruction residuals are systematically larger for non-planetary events.

relative error of 3.2 per cent and the depths 6.5 per cent (Table 5). The Kepler pipeline’s own fits, which use the unbinned light curve and a full physical model, do better at 2.0 and 3.3 per cent.

We do not claim the decoder as a better parameter estimator; it is not, and it was not trained to be. The result establishes something else, namely that the physical bottleneck is genuinely physical. A latent space that reproduces published durations to a few per cent without ever being shown a duration is encoding transit geometry rather than an arbitrary code, which is what licenses reading its ingress-softness parameter as a U-versus-V discriminant (Figure 5).

6.6. Zero-shot transfer to TESS

Applied without any adaptation to 2,351 TESS objects of interest (1,298 confirmed or known planets, 1,053 false positives or false alarms), both net-

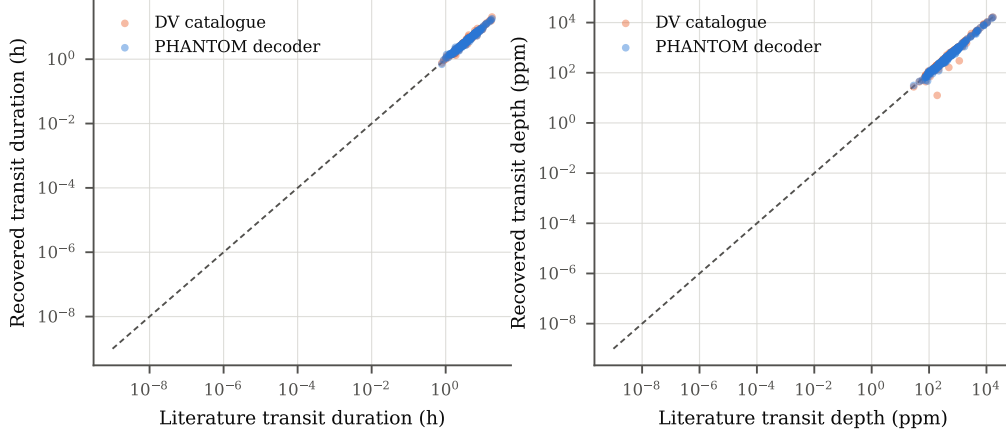


Figure 6: Recovered transit durations and depths against published values for the 264 confirmed planets in the test split, compared with the Kepler pipeline fit.

works degrade sharply: PHANTOM reaches an average precision of 0.789 (AUC 0.751) and ASTRONET 0.797 (AUC 0.767), against a chance rate of 0.552. Both retain real discriminative power, and neither is usable as a TESS vetter at Kepler-level accuracy.

The two models are statistically indistinguishable here, so the architectural advantage established on Kepler does not survive the mission transfer. This corroborates Kopparapu et al. (2026), who report that single-mission training transfers poorly across instruments and recommend joint training. The causes are not mysterious: TESS light curves are two-minute rather than 29.4-minute cadence, span four sectors rather than four years, have a pixel scale an order of magnitude coarser, and the TOI population is dominated by shorter-period, brighter-host systems than the Kepler TCE population. We report the result as a bound on what zero-shot transfer buys, not as a negative verdict on either architecture.

7. Discussion

7.1. What this study does and does not establish

The reliable finding is a comparative one: on a fixed benchmark, with matched training budget and identical splits, a multi-channel architecture with per-view tokens beats the standard two-view convolutional baseline by a margin that is large in operational terms — most of a factor of three in

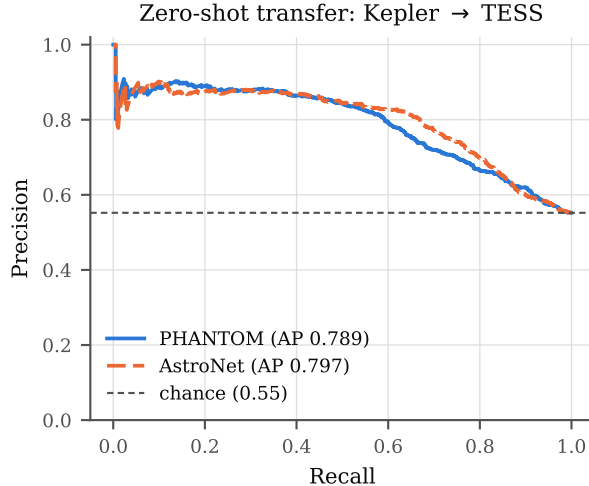


Figure 7: Zero-shot transfer to 2,351 labelled TESS objects of interest. Both architectures retain discriminative power well above the chance rate but are far from their Kepler accuracy, and are indistinguishable from one another.

wasted follow-up at fixed recall — and does so with fewer parameters. That result is stable across seeds and split protocols.

What the study does not establish is *why*, at the level of individual components. We set out to demonstrate that explicit harmonic reasoning would resolve the half-period alias that defeats the classical odd-even test, and built both the harmonic views and a contrast head (Eq. 1) to that end. Neither survives ablation: the contrast head is worth $-0.0007 \pm \text{noise}$, and the harmonic views $+0.0030$. We report this because the alternative — quietly dropping the ablation and keeping the mechanism in the architecture diagram — is how unfalsifiable folklore accumulates in an applied field.

The honest reading is that the harmonic evidence is real but redundant. A binary detected at its half-period alias is visible in the $2P$ fold, but it is also visible in the odd-even depth ratio, in the secondary-phase view and in the decoder’s shape parameter. Supplying it a fourth time adds little. This is consistent with the monotone degradation in Table 3 and with the fact that the full gap to ASTRONET only appears once the architecture changes as well.

7.2. The strength of the tabular baseline

That gradient-boosted trees on summary statistics match a deep multi-channel network on average precision is, we think, the most practically important observation here. The Kepler data-validation products already encode much of what vetting requires, and a strong tabular learner extracts it cheaply and interpretably. The deep model earns its keep specifically in the high-recall regime and in settings where those diagnostics are unavailable — which includes, notably, any survey without a mature data-validation pipeline behind it. Reporting a deep model against only ASTRONET and the classical cascade, as is common, overstates the case for depth.

7.3. Limitations

Our labels are Autovetter dispositions (McCauliff et al., 2015), themselves the output of an automated classifier, so the ceiling reported here is agreement with that labelling rather than with ground truth; the PC class in particular contains an unknown fraction of unconfirmed candidates. The comparison to ASTRONET is to our reimplementations on our preprocessing chain, not to the original authors’ trained model. The conformal analysis assumes exchangeability between validation and test events, which the star-level split makes plausible but does not guarantee, and it is calibrated on a set of $\sim 1,200$ negatives, which as shown bounds the smallest usable target rate. The TESS experiment uses at most four sectors per target and TFOPWG dispositions that are themselves provisional. Finally, five seeds bound our resolution: an ablation worth 0.005 in average precision would not be reliably detectable here, so our null results exclude large effects, not small ones.

8. Conclusions

We presented PHANTOM, a transit-vetting network that combines nine phase-folded views — including folds at competing period hypotheses — through a transformer over per-view tokens, together with a differentiable transit model fitted by reconstruction alone. On the Kepler DR24 benchmark it improves average precision over a matched ASTRONET baseline from 0.9385 to 0.9732 and cuts the false-positive content of a 95 per cent recall candidate list by 62 per cent, using fewer parameters. Conformal p -values with Benjamini–Hochberg selection deliver false-discovery-rate control at the five-to-twenty per cent targets a follow-up programme operates in, and the

unsupervised decoder recovers transit durations and depths to a few per cent of published values while providing an interpretable U-versus-V shape discriminant.

We also report what did not work. No individual component of the architecture is statistically responsible for its advantage; the star-level split we adopted to prevent leakage makes no measurable difference; gradient-boosted trees on summary statistics match the network on average precision; conformal control fails below a five per cent target; and zero-shot transfer to TESS degrades both architectures to indistinguishable performance. Each of these is a result a practitioner benefits from knowing, and none of them would have appeared in a paper reporting only its successes.

The broader methodological point concerns attribution under redundancy. Vetting is a problem in which several physically distinct diagnostics point at the same false positives, and in which the standard leave-one-out ablation therefore reports that nothing matters while the ensemble plainly does. Future work on architectures for this task should report nested ablation sequences rather than single deletions, and should include a strong tabular baseline. We release the full pipeline — acquisition, preprocessing, training, evaluation and figure generation — to make both straightforward.

Data availability

All photometry and catalogues used here are public: Kepler and TESS light curves from the Mikulski Archive for Space Telescopes, and the DR24 TCE, cumulative KOI, TOI and composite planetary parameter tables from the NASA Exoplanet Archive. Code to reproduce every figure and table is released with this paper.

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